

TICVAI

ACCREDITATION MODULE

Functional Design & Board Reference

A structured reference covering command-center operations, holder identity, approvals, credentials, access rights, lifecycle management, communications, analytics, integration and audit.

PROFESSIONAL EDITION

TABLE OF CONTENTS

Board 1 — Accreditation Command Center & Registration.....	3
Board 2 — Accreditation Holder Profile & Identity Management.....	11
Board 3 — Accreditation Approval Workflow & Governance.....	21
Board 4 — Credential Issuance, Media & Badge Management.....	29
Board 5 — Access Rights, Zones & Schedule Assignment.....	38
Board 6 — Accreditation Lifecycle, Validity & Renewal Management.....	47
Board 7 — Notifications, Communications & Bulk Operations.....	57
Board 8 — Analytics, Reporting, Integration & Audit.....	67

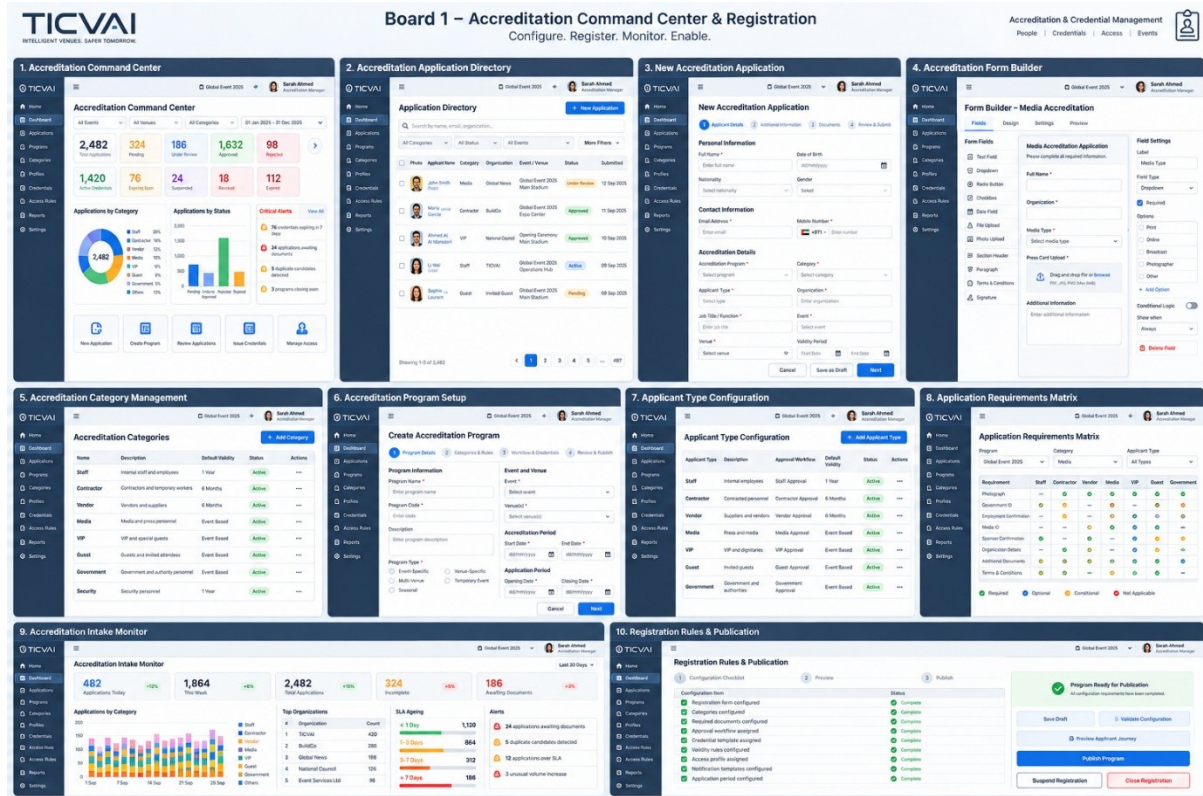
Each board contains ten detailed screen sections covering purpose, scope, operational functions, requirements and use cases.

BOARD 1 — ACCREDITATION COMMAND CENTER & REGISTRATION

Board Objective

Provide the central operational workspace used by accreditation teams to configure accreditation programs, receive applications, categorize applicants, monitor volumes and initiate the accreditation lifecycle.

This board establishes the entry point for **staff, contractors, vendors, media, VIPs, guests and government/authority representatives**.



Screen 1 — Accreditation Command Center

Purpose: Executive and operational landing screen for the accreditation module.

Scope of Work:

The system shall provide a centralized Accreditation Command Center showing the current state of accreditation operations across the tenant.

The screen shall display configurable KPI cards including:

- Total applications
- Pending applications
- Under review
- Approved
- Rejected
- Active credentials
- Expiring soon

- Suspended
- Revoked
- Expired

The dashboard shall allow filtering by tenant, venue, event, accreditation program, category, organization and date.

It shall provide visual summaries of applications by category and accreditation status.

Quick-action buttons shall include:

New Application · Create Program · Review Applications · Issue Credentials · Manage Access

The system shall provide alerts for accreditation exceptions requiring operational attention.

Key requirements: 12.1.1, 12.1.47

Screen 2 — Accreditation Application Directory

Purpose: Central repository of all accreditation applications.

Scope of Work:

The system shall provide a searchable and filterable application directory.

Each application record shall display:

- Application ID
- Applicant name
- Photograph
- Accreditation category
- Organization/company
- Event
- Venue
- Submission date
- Current status
- Assigned reviewer
- Credential status
- Expiry date

Users shall be able to search by applicant name, email, mobile number, organization, accreditation number or credential ID.

Bulk operational actions shall be supported subject to RBAC permissions.

Status indicators shall clearly distinguish:

Draft / Submitted / Under Review / Additional Information Required / Approved / Rejected / Suspended / Revoked / Expired

Screen 3 — New Accreditation Application

Purpose: Allow authorized backend personnel to manually create accreditation applications.

Scope of Work:

The system shall provide a guided accreditation registration interface.

Operators shall be able to select:

- Accreditation program
- Accreditation category
- Applicant type
- Event
- Venue
- Validity period
- Organization

Applicant information may include:

- Full name
- Identification information
- Contact details
- Organization
- Job title/function
- Nationality where permitted
- Emergency information where configured
- Photograph
- Supporting documents

The system shall validate mandatory fields and configured eligibility requirements before submission.

Key requirements: 12.1.2, 12.1.16–12.1.19

Screen 4 — Accreditation Form Builder

Purpose: Configure applicant-facing registration forms without software development.

Scope of Work:

The system shall provide a configurable drag-and-drop Accreditation Form Builder.

Administrators shall be able to configure:

- Text fields
- Dropdowns
- Radio buttons
- Checkboxes
- Date fields
- Document uploads

- Photograph uploads
- Identification fields
- Organization fields
- Conditional questions
- Terms and declarations
- Consent fields

Fields may be configured as:

Mandatory / Optional / Conditional / Hidden

Conditional logic shall allow questions or documents to appear based on applicant category, organization, event or responses.

Form versions shall be maintained for audit purposes.

Key requirement: 12.1.2

Screen 5 — Accreditation Category Management

Purpose: Configure reusable accreditation categories.

Scope of Work:

The system shall support creation and administration of accreditation categories such as:

- Staff
- Contractor
- Vendor
- Media
- VIP
- Guest
- Government / Authority
- Security
- Medical
- Operations
- Sponsor
- Production Crew

Each category may define defaults for:

- Registration form
- Approval workflow
- Credential design
- Access profile
- Validity
- Required documents
- Identity verification

- Notification rules

Categories shall support active/inactive status, effective dates and versioning.

Key requirements: 12.1.8–12.1.15

Screen 6 — Accreditation Program Setup

Purpose: Create the accreditation program governing a particular event, venue, season or organization.

Scope of Work:

Administrators shall be able to create accreditation programs and define:

- Program name
- Program code
- Tenant
- Event
- Venue/venues
- Accreditation period
- Application opening date
- Application closing date
- Allowed categories
- Registration channel
- Maximum application volumes where applicable
- Approval workflow
- Credential template
- Default access profile

Programs shall support:

Event-specific / Venue-specific / Multi-venue / Temporary / Seasonal accreditation.

Key requirements: 12.1.38–12.1.42

Screen 7 — Applicant Type Configuration

Purpose: Configure operational rules according to the type of person being accredited.

Rather than building completely separate systems for staff, contractors, media, VIPs, etc., TICVAI should provide **one accreditation engine with configurable applicant types**.

For each type, administrators shall define:

- Mandatory information
- Required documents
- Identity verification requirement
- Sponsor requirement

- Organization requirement
- Approval path
- Default validity
- Credential type
- Default access rules
- Renewal eligibility

This architecture significantly reduces duplicated configuration.

Key requirements: 12.1.9–12.1.15

Screen 8 — Application Requirements Matrix

Purpose: Configure what an applicant must provide before an application can progress.

The system shall allow administrators to configure requirements by:

Program × Category × Applicant Type

Requirements may include:

- Photograph
- Passport/ID
- Employment confirmation
- Contractor authorization
- Media identification
- Government identification
- Sponsor confirmation
- Vehicle information
- Additional supporting documents

Each requirement may be:

Required / Optional / Conditional / Not Applicable

Missing requirements shall prevent submission or approval according to configuration.

Screen 9 — Accreditation Intake Monitor

Purpose: Give operations teams visibility into incoming accreditation demand.

The screen shall show:

- Applications today
- Applications this week
- Application volumes by category
- Application volumes by organization
- Incomplete applications
- Applications awaiting documents

- Duplicate candidates
- Applications requiring review
- SLA ageing

Users shall be able to drill directly into individual application records.

Operational alerts shall highlight abnormal application volumes or processing bottlenecks.

Screen 10 — Registration Rules & Publication

Purpose: Final governance screen before opening an accreditation program for applications.

The system shall provide a validation checklist covering:

- Registration form configured
- Categories configured
- Required documents configured
- Approval workflow assigned
- Credential template assigned
- Validity rules configured
- Access profile assigned
- Notification templates configured
- Application period configured

Administrators shall be able to:

Save Draft · Validate Configuration · Preview Applicant Journey · Publish · Suspend Registration · Close Registration

Publication and configuration changes shall be captured in the audit trail.

Board 1 Main Acceptance Criteria

Board 1 shall be considered complete when authorized TICVAI users can configure a complete accreditation program without development intervention, including registration forms, accreditation categories, applicant types, required documentation and program rules.

The solution must support all primary accreditation populations—**staff, contractors, vendors, media, VIPs, guests and government/authority personnel**—through a common configurable engine rather than duplicated workflows.

All configuration changes shall be tenant-aware, role-controlled, versioned and auditable.

Important architecture decision

For the later boards, I recommend that we **do not replicate Gate Configuration, QR validation, anti-passback, turnstile configuration or scanner management**, because we already designed these within the **Access Control module**.

Accreditation will instead say:

Person → Accreditation → Credential → Access Profile → Zones/Times → Access Control Engine

BOARD 2 — ACCREDITATION HOLDER PROFILE & IDENTITY MANAGEMENT

Board Objective

Provide a complete operational workspace to create, review, verify, maintain and audit accreditation holder profiles before any credential is approved or issued.

This board shall manage the individual's identity, organization relationship, photographs, documents, verification status, duplicate detection, profile history and compliance readiness.

The accreditation holder profile shall act as the **single source of truth** for the person across accreditation programs, events and venues.



Screen 1 — Accreditation Holder Directory

Purpose: Central searchable directory of all accredited holders.

Scope of Work

The system shall provide a consolidated directory of all persons registered within the accreditation module.

Each holder record shall display:

- Holder ID
- Profile photograph
- Full name
- Accreditation category

- Applicant type
- Organization/company
- Current accreditation status
- Identity verification status
- Active credentials
- Linked event/venue
- Expiry status
- Last updated date

Users shall be able to search by:

- Name
- Holder ID
- Email
- Mobile
- Passport/ID number where permitted
- Organization
- Accreditation ID
- Credential ID

The screen shall support filtering by:

- Category
- Applicant type
- Event
- Venue
- Organization
- Verification status
- Accreditation status
- Credential status

Authorized users shall be able to open, edit, suspend or review profiles according to RBAC permissions.

Screen 2 — Accreditation Holder Profile

Purpose: Master profile screen for an accredited person.

Scope of Work

The system shall maintain a detailed holder profile including:

- Full legal name
- Preferred/display name
- Date of birth
- Nationality where applicable

- Gender where required
- Contact details
- Address where configured
- Job title/function
- Organization
- Department
- Applicant type
- Accreditation category
- Emergency contact where configured
- Internal notes
- External reference IDs

The profile shall display summarized information for:

- Current accreditations
- Credential status
- Access rights
- Verification status
- Documents
- Events and venues
- Renewal history
- Audit history

The holder profile shall remain persistent even when individual accreditations expire.

Key requirement: 12.1.16

Screen 3 — Identity Details & Verification

Purpose: Capture identity data and verify the holder before approval.

Scope of Work

The system shall support identity verification prior to accreditation approval.

The screen shall capture configurable identity data such as:

- Passport number
- National ID
- Emirates ID or equivalent
- Issuing country
- Issue date
- Expiry date
- Date of birth
- Legal name

- Verification source
- Verification reference

Verification statuses shall include:

Not Started / Pending / Verified / Failed / Manual Review / Expired

Verification may be completed through:

- Manual administrator review
- Third-party identity verification integration
- Document validation service
- External government/authority API where applicable

The system shall record:

- Verified by
- Verification date/time
- Verification method
- Outcome
- Reason for failure or exception

Key requirement: 12.1.19

Screen 4 — Photo Management

Purpose: Manage accreditation holder photographs.

Scope of Work

The system shall allow users to:

- Upload a photograph
- Capture a photo from an authorized camera
- Replace an existing photo
- Preview the image
- Crop and position for badge use
- Validate minimum image requirements
- Mark photo as approved/rejected

The system shall support configurable photo rules including:

- Accepted file format
- Minimum resolution
- Maximum file size
- Background requirements
- Face visibility requirements
- Recency rules

Photo records shall include:

- Upload date
- Uploaded by
- Approval status
- Approval user
- Previous photo history

Where facial recognition is enabled in other TICVAI modules, the profile shall be able to reference the approved identity photograph without duplicating image storage unnecessarily.

Key requirement: 12.1.17

Screen 5 — Document Repository

Purpose: Store and manage accreditation-related supporting documents.

Scope of Work

The system shall provide a secure document repository for each accreditation holder.

Supported documents may include:

- Passport
- National ID
- Employment letter
- Contractor authorization
- Vendor registration
- Media identification
- Government authorization
- Sponsor letter
- Insurance certificates
- Background-check documentation
- Custom supporting documents

Each document shall maintain:

- Document type
- File name
- Issue date
- Expiry date
- Verification status
- Uploaded by
- Upload date/time
- Notes
- Version history

The system shall enforce tenant-level security and role permissions for viewing or downloading sensitive files.

Key requirement: 12.1.18

Screen 6 — Document Verification Queue

Purpose: Operational review queue for uploaded accreditation documentation.

Scope of Work

The system shall provide a configurable document review workflow.

Reviewers shall see:

- Holder name
- Document type
- Related accreditation
- Submission date
- Expiry date
- Current verification status
- Assigned reviewer
- SLA ageing

Review actions shall include:

Approve / Reject / Request Replacement / Escalate / Add Comment

Rejection shall require a reason code and optional reviewer notes.

Approved documentation shall be marked as verified and become eligible for accreditation approval workflows.

The system shall support bulk review actions where permitted.

Screen 7 — Duplicate & Identity Conflict Detection

Purpose: Prevent duplicate holder records and potential identity conflicts.

Scope of Work

The system shall identify possible duplicates based on configurable matching logic such as:

- Full name
- Date of birth
- Email
- Mobile
- Passport/ID number
- Organization
- Photograph or face-match reference where legally enabled

Potential duplicates shall be shown with a confidence indicator.

Reviewers shall be able to:

- Confirm duplicate

- Merge records
- Link records
- Dismiss match
- Escalate for investigation

The system shall prevent accidental duplicate credential issuance where configured.

All merges and identity-resolution decisions shall be fully auditable.

Screen 8 — Organization & Affiliation Management

Purpose: Link holders to their employer, contractor, media organization, government authority or sponsor.

Scope of Work

The system shall support association of accreditation holders with one or more organizations.

Organization information may include:

- Organization name
- Organization type
- Registration/reference number
- Primary contact
- Sponsor contact
- Contract/reference
- Validity dates
- Status

Users shall be able to view all accreditation holders linked to the same organization.

The system shall support organization types such as:

- Internal staff
- Contractor
- Vendor
- Media organization
- Government authority
- Sponsor
- Partner
- Production company

Organization relationships shall be usable later in approval workflows and access-right assignment.

Screen 9 — Profile Completeness & Compliance Monitor

Purpose: Show whether a holder profile is ready to proceed to accreditation approval.

Scope of Work

The system shall calculate and display profile completeness based on configured program requirements.

The readiness checklist may include:

- Personal information complete
- Valid photograph
- Mandatory documents uploaded
- Mandatory documents verified
- Identity verification completed
- Organization confirmed
- Consent/declarations accepted
- No unresolved duplicate alert
- No expired mandatory document

The system shall visually indicate:

Ready / Incomplete / Attention Required / Blocked

Administrators shall be able to drill down into missing or failed requirements.

This screen shall help accreditation teams resolve issues before moving applications into approval.

Screen 10 — Profile History & Audit Timeline

Purpose: Provide a complete historical record of holder profile changes.

Scope of Work

The system shall maintain a detailed chronological timeline of profile activity, including:

- Profile creation
- Personal detail changes
- Photograph updates
- Document uploads
- Document approval/rejection
- Identity verification
- Organization changes
- Duplicate merge actions
- Accreditation associations
- Status changes

Each audit event shall capture:

- User

- Date/time
- Action
- Previous value
- New value
- Source/channel
- Related accreditation or event

The audit history shall be read-only for normal users and shall support export for authorized compliance personnel.

Board 2 Acceptance Criteria

Board 2 shall be considered complete when TICVAI can maintain a **single verified accreditation-holder profile** that supports multiple accreditation programs, venues and events without recreating the same person repeatedly.

The solution must support secure management of photographs and accreditation-related documents, identity verification before approval, organization relationships, duplicate detection, profile readiness checks and full profile history.

All sensitive data shall be protected through **RBAC, tenant segregation, audit logging and configurable retention policies**.

Requirement Coverage

This board directly covers:

12.1.16 — Profile Management

12.1.17 — Photo Management

12.1.18 — Document Management

12.1.19 — Identity Verification

It also provides supporting capability for later requirements covering **approval**,

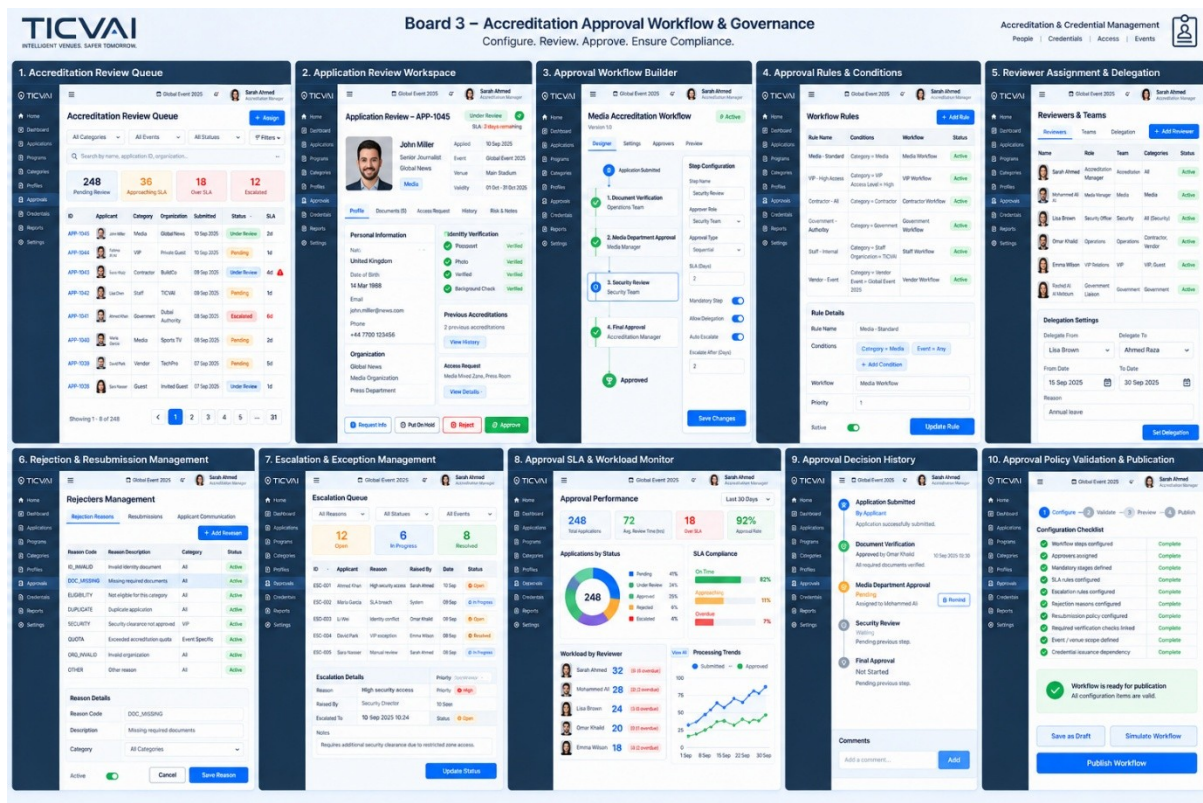
BOARD 3 — ACCREDITATION APPROVAL WORKFLOW & GOVERNANCE

Board Objective

Provide a configurable workflow engine for accreditation review, approval, rejection, escalation and resubmission.

The board shall ensure that no accreditation is approved unless required identity, document, profile and policy checks have been completed.

It shall support different approval paths by accreditation category, event, venue, organization and risk level.



Screen 1 — Accreditation Review Queue

Purpose: Central operational queue for applications requiring review.

Scope of Work

The system shall display all accreditation applications awaiting review.

Each record shall show:

- Application ID
- Applicant name
- Photo
- Accreditation category
- Organization

- Event / venue
- Submission date
- Verification status
- Document status
- Current workflow stage
- Assigned reviewer
- SLA ageing
- Risk / exception indicator

Filters shall include:

- Category
- Event
- Venue
- Organization
- Reviewer
- Workflow stage
- SLA status
- Verification status

Actions shall include:

Open Review / Assign Reviewer / Reassign / Escalate / Bulk Assign

Key requirement: 12.1.3

Screen 2 — Application Review Workspace

Purpose: Give reviewers a complete decision workspace.

Scope of Work

The screen shall present all information required to approve or reject an accreditation application.

Reviewers shall be able to view:

- Applicant profile
- Photo
- Identity verification status
- Submitted documents
- Organization affiliation
- Accreditation category
- Requested event / venue
- Requested validity
- Proposed access profile
- Previous accreditations

- Previous rejection or suspension history
- Comments and workflow history

Decision actions shall include:

Approve / Reject / Request More Information / Escalate / Put On Hold

The system shall prevent approval if mandatory conditions remain incomplete.

Screen 3 — Approval Workflow Builder

Purpose: Configure reusable accreditation approval workflows.

Scope of Work

Administrators shall be able to create multi-step workflows using a visual workflow builder.

Workflow stages may include:

- Initial validation
- Identity verification
- Document review
- Department approval
- Security approval
- Event approval
- Venue approval
- Final accreditation approval

Each step shall support:

- Required approver role
- Specific user or team
- Sequential approval
- Parallel approval
- Conditional approval
- Mandatory / optional stage
- SLA target
- Escalation path
- Auto-approval conditions where permitted

The workflow shall be versioned and auditable.

Key requirement: 12.1.3

Screen 4 — Approval Rules & Conditions

Purpose: Define when a particular approval workflow is applied.

Scope of Work

The system shall allow configuration of rules based on:

- Accreditation category
- Applicant type
- Event
- Venue
- Organization
- Government / authority category
- Risk classification
- Access level
- Requested zones
- Validity duration
- Credential type

Example:

Media Accreditation + Backstage Zone → Media Manager Approval + Security Approval

VIP Accreditation + Public Area Only → VIP Relations Approval

Rules shall support priority and conflict resolution.

Screen 5 — Reviewer Assignment & Delegation

Purpose: Manage reviewers and approval responsibilities.

Scope of Work

Administrators shall be able to configure:

- Review teams
- Individual reviewers
- Reviewer roles
- Category ownership
- Venue ownership
- Event ownership
- Approval limits
- Temporary delegation
- Backup reviewer
- Out-of-office delegation

The system shall automatically route applications based on configured responsibilities.

Manual reassignment shall require appropriate permission and be captured in the audit log.

Screen 6 — Rejection & Resubmission Management

Purpose: Control rejected applications and resubmission workflows.

Scope of Work

The system shall support configurable rejection reasons such as:

- Invalid identity document
- Missing documentation
- Failed identity verification
- Eligibility not met
- Duplicate application
- Security rejection
- Accreditation quota exceeded
- Ineligible organization
- Other

Reviewers shall be able to provide internal comments and applicant-visible explanations.

The applicant may be allowed to:

- Correct information
- Replace documents
- Upload additional evidence
- Resubmit the application

The system shall preserve the original submission and all subsequent versions.

Key requirement: 12.1.33

Screen 7 — Escalation & Exception Management

Purpose: Handle applications requiring special review.

Scope of Work

The system shall allow applications to be escalated due to:

- SLA breach
- Identity conflict
- High-security access
- Government / authority status
- VIP exception
- Missing mandatory information
- Manual policy override
- Senior management decision

Each escalation shall include:

- Reason

- Raised by
- Escalated to
- Priority
- Due date
- Resolution
- Supporting comments

Exception approvals shall require enhanced audit logging.

Screen 8 — Approval SLA & Workload Monitor

Purpose: Monitor review performance and bottlenecks.

Scope of Work

The dashboard shall display:

- Pending approvals
- Average review time
- Applications approaching SLA
- Applications over SLA
- Workload by reviewer
- Workload by team
- Approval rate
- Rejection rate
- Escalation volume
- Resubmission volume

Managers shall be able to drill into delayed applications and reassign workloads.

Screen 9 — Approval Decision History

Purpose: Maintain a complete record of every approval decision.

Scope of Work

The system shall record:

- Application
- Workflow stage
- Reviewer
- Decision
- Date/time
- Comments
- Rejection reason
- Escalation reason

- Changed fields
- Supporting documents
- Workflow version

Decision records shall be immutable for standard users.

The full approval path shall be visible as a chronological timeline.

Screen 10 — Approval Policy Validation & Publication

Purpose: Validate and publish approval workflow configurations.

Scope of Work

Before a workflow becomes active, the system shall validate:

- Workflow steps configured
- Approvers assigned
- Mandatory stages defined
- SLA rules configured
- Escalation rules configured
- Rejection reasons configured
- Resubmission policy configured
- Required verification checks linked
- Event / venue scope defined
- Credential issuance dependency configured

Administrators shall be able to:

Save Draft / Validate / Simulate Workflow / Preview Routing / Publish / Deactivate / Clone

Published workflows shall be version-controlled.

Board 3 Acceptance Criteria

Board 3 shall be considered complete when TICVAI can configure and operate accreditation approval workflows without software development.

The solution must support:

- Multi-level approvals
- Conditional routing
- Reviewer assignment
- Rejection management
- Resubmission
- Escalation
- SLA monitoring
- Approval history
- Workflow versioning

- Controlled publication

The system shall not allow credential issuance until the applicable accreditation workflow reaches an approved state.

Requirement Coverage

This board directly covers:

12.1.3 — Approval Workflow

12.1.33 — Accreditation Rejection Management

12.1.44 — Approval Notifications partly, with notification delivery configured later in Board 7.

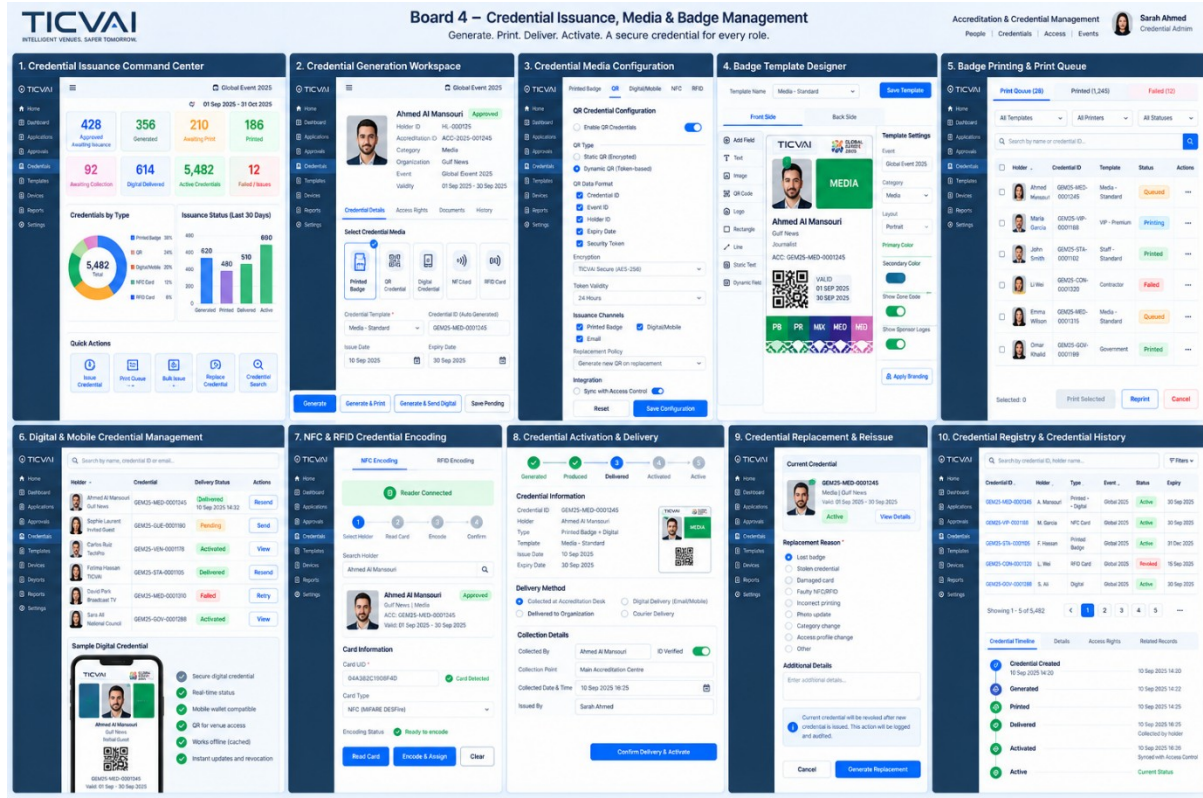
It also supports governance for requirements related to identity verification, credential issuance, suspension, revocation and audit logging.

BOARD 4 — CREDENTIAL ISSUANCE, MEDIA & BADGE MANAGEMENT

Board Objective

Provide a centralized credential-management workspace to generate, personalize, issue, print, activate, replace and manage accreditation credentials.

The solution shall support:



Printed Badge · QR Credential · Mobile/Digital Credential · NFC Card · RFID Card

A single accreditation holder may have one or more authorized credential media while maintaining one underlying accreditation identity and access entitlement.

Screen 1 — Credential Issuance Command Center

Purpose: Central operational dashboard for accreditation credential issuance.

Scope of Work

The system shall provide real-time visibility of:

- Approved applications awaiting credential issuance
- Credentials generated
- Credentials awaiting printing
- Credentials printed
- Credentials awaiting collection
- Digital credentials delivered

- Active credentials
- Failed issuance
- Replacement requests
- Revoked credentials

Users shall be able to filter by:

- Event
- Venue
- Accreditation program
- Category
- Organization
- Credential type
- Issuance status

Quick actions shall include:

Issue Credential / Print Queue / Bulk Issue / Replace Credential / Credential Search

Key requirements: 12.1.4, 12.1.20–12.1.24

Screen 2 — Credential Generation Workspace

Purpose: Convert an approved accreditation into an operational credential.

Scope of Work

The system shall retrieve the approved accreditation holder and display:

- Holder name
- Photograph
- Holder ID
- Accreditation ID
- Category
- Organization
- Event
- Venue
- Validity period
- Assigned access profile

The operator shall select the permitted credential media:

- Printed badge
- QR
- Digital/mobile
- NFC
- RFID

The system shall generate a unique credential identifier.

Credential generation shall only be permitted when the accreditation status is eligible for issuance.

Actions:

Generate / Generate & Print / Generate & Send Digital / Save Pending

Key requirement: 12.1.4

Screen 3 — Credential Media Configuration

Purpose: Configure the credential technologies available for each accreditation program.

Scope of Work

Administrators shall be able to enable and configure:

Physical Badge

Printed accreditation credential.

QR

Static or secure/dynamic QR configuration according to the TICVAI security architecture.

Digital Credential

Mobile-accessible accreditation credential.

NFC

NFC credential associated with supported credential media.

RFID

RFID credential associated with supported cards/tags.

Each media type shall support configuration for:

- Active/inactive
- Credential identifier format
- Activation method
- Issuance channel
- Replacement policy
- Expiry behavior
- Access Control integration

Key requirements: 12.1.21–12.1.24

Screen 4 — Badge Template Designer

Purpose: Design physical accreditation badges.

Scope of Work

The system shall provide a visual badge-template designer.

Administrators shall be able to configure:

- Holder photograph

- Full name
- Organization
- Job title/function
- Accreditation category
- Accreditation ID
- Event name
- Venue
- Validity dates
- QR code
- Color coding
- Zone indicators
- Logos
- Sponsor branding
- Security marks
- Custom fields

The designer shall support front and back badge layouts.

Templates may differ by:

- Tenant
- Event
- Venue
- Accreditation category
- Program

Key requirements: 12.1.20, 12.1.56, 12.1.57

Screen 5 — Badge Printing & Print Queue

Purpose: Manage physical accreditation badge production.

Scope of Work

The system shall provide a centralized printing queue showing:

- Holder
- Photograph
- Accreditation ID
- Badge template
- Printer
- Print status
- Number of copies
- Requested by

- Request time

Statuses shall include:

Queued / Printing / Printed / Failed / Reprint Required / Cancelled

Users shall be able to:

- Print individual badges
- Batch print
- Reprint
- Select printer
- Preview badge
- Cancel job

Reprinting may require a reason depending on configured security policies.

Key requirement: 12.1.20

Screen 6 — Digital & Mobile Credential Management

Purpose: Manage credentials delivered electronically to accreditation holders.

Scope of Work

The system shall support issuance of mobile accreditation credentials.

The digital credential shall display configurable information including:

- Holder photograph
- Name
- Accreditation category
- Organization
- Credential ID
- Event
- Validity
- QR credential
- Current credential status

Administrators shall be able to:

- Issue
- Send
- Resend
- Activate
- Deactivate
- Refresh
- Revoke

Delivery status shall be tracked.

The architecture shall allow future integration with supported mobile-wallet technologies where required.

Key requirement: 12.1.21

Screen 7 — NFC & RFID Credential Encoding

Purpose: Associate physical NFC/RFID media with an accreditation.

Scope of Work

The system shall provide an encoding/assignment workspace for supported credential media.

Operators shall be able to:

- Select holder
- Scan/read card
- Capture UID/media identifier
- Assign credential
- Encode supported information
- Validate media
- Activate credential
- Replace existing media

The system shall prevent the same credential media from being assigned to multiple active holders unless explicitly supported by configuration.

The screen shall show:

Reader Connected / Card Detected / Encoding / Verification / Successful / Failed

Key requirements: 12.1.23, 12.1.24

Screen 8 — Credential Activation & Delivery

Purpose: Control when an issued credential becomes operational.

Scope of Work

The system shall track the complete credential fulfillment process:

Generated → Produced → Delivered → Activated → Active

Delivery methods may include:

- Badge collection
- Accreditation desk
- Digital delivery
- Email activation
- Mobile application
- Authorized organization representative

For physical credentials, the system shall capture:

- Collected by
- Collection date/time
- Issued by
- Collection point
- Optional acknowledgment

Activation shall synchronize credential status with the Access Control layer.

Screen 9 — Credential Replacement & Reissue

Purpose: Securely manage lost, damaged, stolen or incorrectly issued credentials.

Scope of Work

Operators shall be able to initiate replacement for:

- Lost badge
- Stolen credential
- Damaged card
- Failed NFC/RFID
- Incorrect printing
- Photo update
- Category change
- Access-profile change
- Other approved reason

Before replacement, the system shall display the existing credential.

The previous credential shall be:

Revoked / Deactivated / Superseded

according to policy.

A new unique credential identifier may then be generated.

All replacement activities shall be auditable.

Screen 10 — Credential Registry & Credential History

Purpose: Maintain the authoritative record of every credential issued by TICVAI.

Scope of Work

The registry shall show:

- Credential ID
- Holder
- Accreditation ID
- Credential/media type
- Event

- Venue
- Category
- Issue date
- Activation date
- Expiry
- Current status
- Replacement relationship

Credential statuses shall include:

Pending / Generated / Printed / Delivered / Active / Suspended / Expired / Revoked / Replaced

Opening a credential shall display its complete lifecycle timeline:

Created → Generated → Printed/Encoded → Delivered → Activated → Modified → Suspended/Revoked/Expired/Replaced

All credential events shall be auditable.

Board 4 Acceptance Criteria

Board 4 shall be considered complete when an approved accreditation can be converted into a secure operational credential through a controlled and auditable issuance process.

TICVAI shall support:

- Physical accreditation badges
- QR credentials
- Digital/mobile credentials
- NFC credentials
- RFID credentials
- Badge design and branding
- Badge printing
- NFC/RFID assignment and encoding
- Credential activation
- Credential delivery
- Replacement/reissue
- Complete credential registry and history

Credential issuance shall be blocked where the underlying accreditation has not completed its required approval workflow.

When an accreditation or credential is subsequently **suspended, revoked or expired**, the status shall propagate to the Access Control layer so the credential can no longer provide unauthorized access.

Matrix Coverage

Board 4 directly addresses:

12.1.4 — Credential Generation

12.1.20 — Credential Printing

12.1.21 — Digital Credentials

12.1.22 — QR Credential Support

12.1.23 — NFC Credential Support

12.1.24 — RFID Credential Support

12.1.56 — Accreditation Templates (*credential-template portion*)

12.1.57 — Accreditation Branding (*credential-branding portion*)

It also prepares the credential for **Board 5 — Access Rights, Zones & Schedule Assignment**, where we connect the approved accreditation credential to exactly **where, when and under what conditions the holder is allowed access**.

BOARD 5 — ACCESS RIGHTS, ZONES & SCHEDULE ASSIGNMENT

Board Objective

Provide a centralized workspace to configure, assign, preview and govern accreditation access entitlements based on **venue, zone, operational area, date, time and schedule**.



The system shall allow access rights to be inherited from accreditation categories or templates while supporting controlled individual exceptions.

The intended architecture is:

Holder → Accreditation → Credential → Access Profile → Venue → Zone / Area → Date & Time Rules → Access Control

Screen 1 — Accreditation Access Command Center

Purpose: Central operational dashboard for accreditation access permissions.

Scope of Work

The system shall provide an overview of accreditation access assignments across events and venues.

Dashboard KPIs shall include:

- Active access profiles
- Accredited holders with access
- Credentials with assigned access
- Restricted-access holders

- Temporary access grants
- Expiring permissions
- Suspended access
- Revoked access
- Access exceptions
- Unassigned approved credentials

Users shall be able to filter by:

- Event
- Venue
- Accreditation program
- Category
- Organization
- Zone
- Access profile
- Credential status

Quick actions:

Create Access Profile / Assign Access / Temporary Grant / Revoke Access / View Access Matrix

Key requirements: 12.1.5, 12.1.25–12.1.30

Screen 2 — Access Profile Management

Purpose: Create reusable accreditation access profiles.

Scope of Work

Administrators shall be able to create access profiles such as:

- Staff – Full Venue
- Contractor – Operational Areas
- Media – Press Areas
- Media – Field Access
- VIP – Hospitality
- Government – Authority Access
- Vendor – Service Areas
- Production – Backstage
- Security – Controlled Zones

Each profile shall define:

- Venue
- Permitted zones
- Permitted operational areas

- Date validity
- Time validity
- Access schedule
- Accreditation categories allowed
- Applicable events
- Default credential behavior

Profiles shall support:

Draft / Active / Suspended / Archived

A profile may be reused across multiple accreditation holders.

Key requirement: 12.1.5

Screen 3 — Venue & Zone Access Matrix

Purpose: Visually configure which accreditation profiles can access each venue zone.

Scope of Work

The system shall provide a matrix showing:

Access Profile × Venue Zone

Example zones may include:

- Public Area
- Staff Area
- Media Centre
- Press Conference Room
- VIP Lounge
- Hospitality
- Backstage
- Field of Play
- Control Room
- Loading Area
- Technical Area
- Restricted Security Area

Administrators shall be able to mark access as:

Allowed / Restricted / Conditional / Not Allowed

The matrix shall support filtering by event and venue.

Changes shall display impact information before publication.

Key requirement: 12.1.25

Screen 4 — Operational Area Permission Management

Purpose: Configure access to operational areas below the broader zone level.

Scope of Work

The system shall support more granular permissions for areas such as:

- Accreditation office
- Broadcast compound
- Press room
- Locker-room corridor
- Production office
- Catering area
- Storage area
- Loading dock
- Technical room
- Security command centre

Each area may be associated with:

- Parent venue
- Parent zone
- Access profile
- Accreditation category
- Operational role
- Schedule

The system shall support inclusion and exclusion rules.

Key requirement: 12.1.26

Screen 5 — Date & Time Access Rules

Purpose: Control when accreditation access is valid.

Scope of Work

Administrators shall be able to configure:

- Valid from date
- Valid until date
- Allowed days
- Access start time
- Access end time
- Event-day restrictions
- Setup-day access
- Breakdown-day access

- Grace period
- Overnight access
- Blackout periods

Example:

Media – Field Access

Venue: Stadium A

Zone: Field of Play

Date: 15 September 2026

Access: 16:00–23:00

Outside the configured period, the credential shall not provide access to that entitlement.

Key requirements: 12.1.27, 12.1.28

Screen 6 — Access Schedule Management

Purpose: Create reusable schedules for accreditation permissions.

Scope of Work

Administrators shall be able to create schedules such as:

Event Day Staff

06:00–23:59

Media

3 hours before event → 2 hours after event

Contractor

08:00–18:00 Monday–Friday

VIP

Event opening → event close

Production Crew

24-hour event-period access

Schedules shall support:

- Recurring rules
- Event-relative schedules
- Fixed schedules
- Date exceptions
- Holiday exceptions
- Special-event overrides

Schedules may be attached to access profiles or individual accreditation records.

Key requirement: 12.1.29

Screen 7 — Holder Access Assignment

Purpose: Assign access permissions to an individual accreditation holder.

Scope of Work

The screen shall display:

- Holder
- Photograph
- Accreditation
- Category
- Credential
- Event
- Venue
- Current access profile

Administrators shall be able to:

- Assign an access profile
- Add permitted zone
- Remove zone
- Add operational area
- Apply schedule
- Apply date/time restriction
- Add temporary permission

The system shall clearly distinguish:

Inherited Access

from the accreditation category/profile.

and

Individual Override

manually granted to the holder.

High-risk overrides may require approval.

Screen 8 — Temporary Access & Exception Management

Purpose: Provide controlled short-term access outside normal accreditation permissions.

Scope of Work

Authorized users shall be able to issue temporary access for:

- Specific zone
- Specific operational area
- Specific date
- Specific time window

- Specific event

Each temporary grant shall require:

- Holder
- Reason
- Requested access
- Start date/time
- End date/time
- Requested by
- Approved by where required

Temporary access shall expire automatically.

Example:

Contractor temporarily granted Control Room access from 14:00–16:00 for maintenance.

All temporary grants shall be auditable.

Screen 9 — Access Revocation & Suspension

Purpose: Immediately remove or temporarily disable accreditation access.

Scope of Work

Authorized users shall be able to:

Suspend Access

Temporarily prevent use while retaining the accreditation.

Revoke Access

Permanently remove selected or all accreditation access rights.

Users shall be able to select:

- All access
- Venue
- Zone
- Area
- Specific entitlement
- Credential

The system shall require:

- Reason
- Effective time
- Operator
- Optional notes

Revocation shall be propagated immediately to the Access Control engine.

For offline access devices, the status shall be included in the next supported synchronization/revocation update mechanism.

Key requirement: 12.1.30

Screen 10 — Access Rights Preview, Impact & Synchronization

Purpose: Validate accreditation access before publishing it to Access Control.

Scope of Work

The system shall provide a human-readable preview showing exactly:

Ahmed Al Mansouri
Media Accreditation
Stadium A
Media Centre — Allowed
Press Room — Allowed
Field of Play — 16:00–23:00 only
VIP Lounge — Not Allowed
Control Room — Not Allowed

The screen shall validate:

- Credential active
- Accreditation active
- Venue assigned
- Zones configured
- Areas configured
- Date validity
- Time validity
- Schedule conflicts
- Expired permissions
- Conflicting access rules

Administrators shall then be able to:

Save Draft / Validate / Preview / Publish Access / Synchronize

The system shall show synchronization status with Access Control.

Board 5 Acceptance Criteria

Board 5 shall be considered complete when TICVAI can determine precisely:

WHO can access WHAT, WHERE, WHEN, and under WHICH accreditation.

Access permissions shall support inheritance through reusable profiles while permitting controlled individual overrides.

The solution must support:

- Venue-based access
- Zone-based access
- Operational-area permissions
- Date-based permissions

- Time-based permissions
- Reusable access schedules
- Individual access assignment
- Temporary access
- Suspension
- Immediate revocation
- Access preview and validation
- Synchronization with Access Control

All changes to access rights shall be recorded in the accreditation audit trail.

Matrix Coverage

Board 5 directly addresses:

12.1.5 — Access Assignment

12.1.25 — Zone-Based Access

12.1.26 — Area-Based Permissions

12.1.27 — Time-Based Access Rights

12.1.28 — Date-Based Access Rights

12.1.29 — Access Schedule Management

12.1.30 — Access Revocation

It also establishes the data required for:

12.1.31 — Access Audit Trail

12.1.49 — Accreditation Access Reporting

Those reporting and historical capabilities will be consolidated later rather than duplicated here.

Important boundary with Access Control

Board 5 **does not configure physical gates, readers, turnstiles, anti-passback, scanner behavior or validation devices**. Those remain in TICVAI's Access Control module.

The relationship is:

Accreditation Board 5:

"Ahmed may enter Media Centre between 16:00–23:00."

Access Control:

"Reader Gate M03 scanned Ahmed's credential at 18:42 → permission valid → ACCESS GRANTED."

BOARD 6 — ACCREDITATION LIFECYCLE, VALIDITY & RENEWAL MANAGEMENT

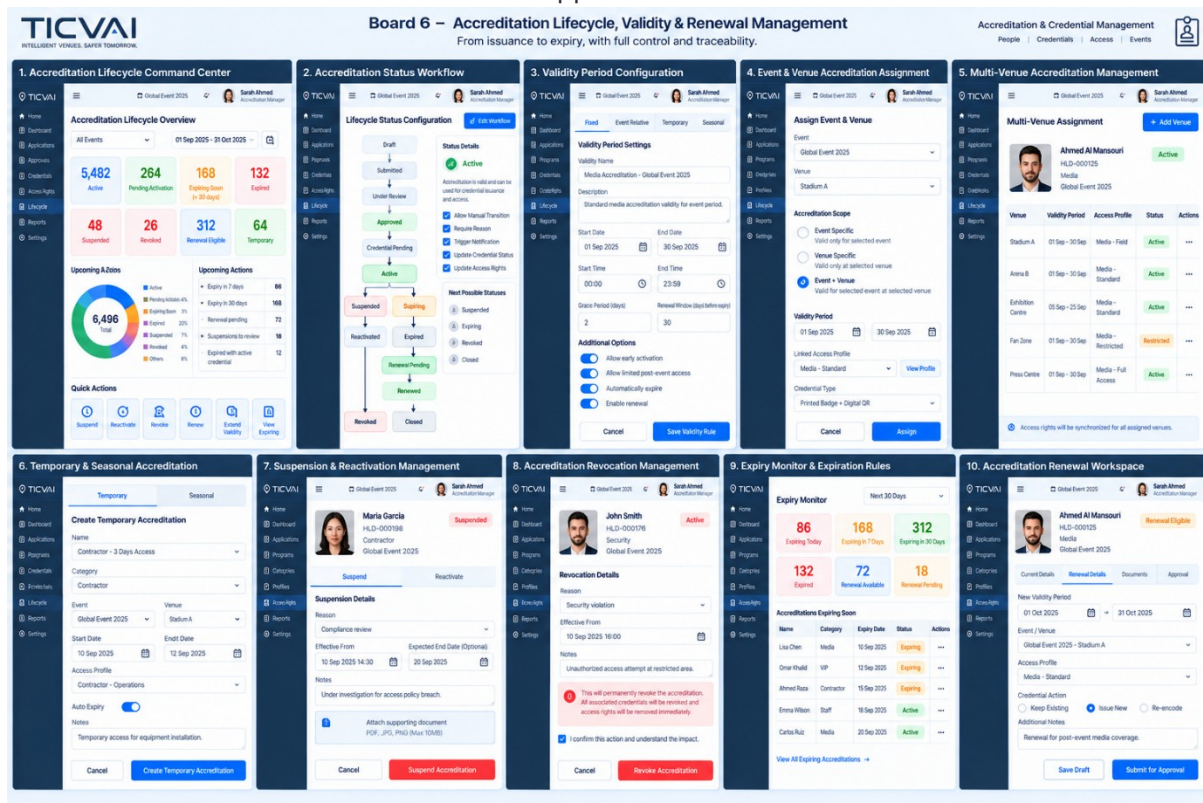
Board Objective

Provide a centralized lifecycle-management capability covering the accreditation from approval and activation through expiry, renewal, suspension, revocation, and closure.

The system shall support:

Approved → Credential Issued → Active → Suspended → Reactivated → Expiring → Renewed / Expired → Revoked / Closed

Lifecycle actions shall automatically propagate to associated credentials and access rights where applicable.



Screen 1 — Accreditation Lifecycle Command Center

Purpose: Central operational dashboard for accreditation lifecycle status.

Scope of Work

The system shall provide KPI cards including:

- Active accreditations
- Pending activation
- Expiring soon
- Expired
- Suspended

- Revoked
- Renewal eligible
- Renewal pending
- Temporary accreditations
- Seasonal accreditations

Users shall be able to filter by:

- Event
- Venue
- Program
- Category
- Organization
- Accreditation type
- Status
- Validity period

The screen shall provide operational alerts for:

- Upcoming expirations
- Pending renewals
- Suspensions requiring review
- Temporary credentials nearing expiry
- Expired accreditation with active credential
- Lifecycle synchronization exceptions

Quick actions:

Suspend / Reactivate / Revoke / Renew / Extend Validity / View Expiring

Key requirements: 12.1.6, 12.1.32, 12.1.34–12.1.37

Screen 2 — Accreditation Status Workflow

Purpose: Configure and manage the accreditation lifecycle statuses.

Scope of Work

The system shall support configurable lifecycle statuses including:

Draft

Submitted

Under Review

Approved

Credential Pending

Active

Suspended

Expiring

Expired

Renewal Pending

Renewed
Revoked
Closed

Administrators shall be able to define permitted transitions.

For example:

Active → Suspended → Active

or

Active → Revoked

but not:

Revoked → Active

unless an authorized exceptional process is configured.

Each transition may define:

- Required permission
- Approval requirement
- Reason requirement
- Notification
- Credential action
- Access-right action

Key requirement: 12.1.32

Screen 3 — Validity Period Configuration

Purpose: Configure how long an accreditation remains valid.

Scope of Work

Administrators shall be able to configure:

- Start date
- End date
- Start time
- End time
- Fixed validity
- Event-relative validity
- Temporary validity
- Seasonal validity
- Permanent accreditation where permitted
- Grace period
- Renewal window

Validity may be inherited from the accreditation program or overridden for an individual holder subject to permissions.

The system shall prevent accreditation use outside its valid period.

Key requirement: 12.1.6

Screen 4 — Event & Venue Accreditation Assignment

Purpose: Determine the event and venue scope of an accreditation.

Scope of Work

The system shall support:

Event-Specific

Valid only for a selected event.

Venue-Specific

Valid only at a selected venue.

Event + Venue

Valid for a specific event at a specific venue.

Administrators shall be able to select:

- Event
- Venue
- Accreditation program
- Validity period
- Applicable credential
- Access profile

Changes shall be checked against existing access assignments.

Key requirements: 12.1.38, 12.1.39

Screen 5 — Multi-Venue Accreditation Management

Purpose: Allow one accreditation to operate across multiple approved venues.

Scope of Work

The system shall allow administrators to assign multiple venues to one accreditation.

For each venue, the system shall display:

- Venue
- Valid from/to
- Access profile
- Permitted zones
- Credential
- Current status

Example:

Regional Operations Staff

- Stadium A — Active
- Arena B — Active
- Exhibition Centre — Active
- Fan Zone — Restricted

Venue permissions may use a common access profile or venue-specific access profiles.

Removing a venue shall synchronize the change with associated access rights.

Key requirement: 12.1.40

Screen 6 — Temporary & Seasonal Accreditation

Purpose: Manage short-duration and recurring accreditation programs.

Scope of Work

The system shall support temporary credentials for:

- One-day events
- Contractors
- Temporary workers
- Media visits
- Special guests
- Vendors
- Maintenance teams
- Event production teams

It shall also support seasonal accreditation for:

- Sports seasons
- Exhibition seasons
- Annual venue operations
- Recurring event programs
- Long-term contractors

Administrators shall configure:

- Accreditation type
- Start/end date
- Events
- Venues
- Access profile
- Automatic expiry
- Renewal eligibility

Temporary accreditations shall automatically expire without manual intervention.

Key requirements: 12.1.41, 12.1.42

Screen 7 — Suspension & Reactivation Management

Purpose: Temporarily disable accreditation without permanently terminating it.

Scope of Work

Authorized users shall be able to suspend an accreditation for reasons including:

- Compliance review
- Security investigation
- Employment status
- Contractor issue
- Missing/expired document
- Temporary restriction
- Operational decision

The operator shall specify:

- Suspension reason
- Effective date/time
- Expected end date where applicable
- Notes
- Supporting evidence
- Approver where required

The system shall define the impact on:

Credential → Suspended

and

Access Rights → Disabled

where configured.

Authorized users may subsequently reactivate the accreditation after required conditions are satisfied.

Key requirement: 12.1.34

Screen 8 — Accreditation Revocation Management

Purpose: Permanently terminate an accreditation.

Scope of Work

Authorized users shall be able to permanently revoke an accreditation.

Revocation reasons may include:

- Security violation
- Accreditation misuse
- Employment termination
- Contractor termination

- Fraud
- Credential sharing
- Policy breach
- Management decision
- Other configured reason

Before revocation, the system shall display the impact:

Accreditation → Revoked

Credential → Revoked

Access Rights → Removed/Blocked

The operator shall confirm the action before execution.

Revocation shall propagate to the Access Control layer without requiring manual deactivation of each associated entitlement.

The system shall retain the accreditation history after revocation.

Key requirement: 12.1.35

Screen 9 — Expiry Monitor & Expiration Rules

Purpose: Proactively manage accreditations approaching expiry.

Scope of Work

The screen shall show accreditation buckets such as:

- Expiring today
- Expiring within 7 days
- Expiring within 30 days
- Expired today
- Recently expired
- Renewal available
- Renewal already initiated

Administrators shall configure expiration behavior including:

- Automatic status change
- Credential deactivation
- Access-right termination
- Grace period
- Renewal eligibility
- Notification trigger

The system shall automatically transition eligible records to **Expired** once validity ends.

Expiry notifications shall connect to Board 7's notification engine.

Key requirements: 12.1.36, 12.1.46

Screen 10 — Accreditation Renewal Workspace

Purpose: Manage accreditation renewal without unnecessarily recreating the holder or application.

Scope of Work

The system shall support renewal of eligible accreditation records.

The renewal workspace shall compare:

Current	Renewal
Category	Proposed Category
Organization	Organization
Event/Venue	New Event/Venue
Validity	New Validity
Credential	Credential Action
Access Profile	Proposed Access
Documents	Current/Expired
Identity Verification	Valid/Reverification Required

Renewal rules may require:

- Updated photograph
- New/updated documents
- Identity reverification
- Organization confirmation
- Manager approval
- Security approval
- Access-right review

Administrators shall be able to:

Start Renewal / Request Updates / Submit for Approval / Approve Renewal / Issue New Credential

The system shall retain a relationship between the previous accreditation and renewed accreditation.

Key requirements: 12.1.37, 12.1.45

Board 6 Acceptance Criteria

Board 6 shall be considered complete when TICVAI can manage an accreditation throughout its full operational lifecycle without deleting or recreating historical records.

The solution shall support:

- Configurable lifecycle statuses
- Validity periods
- Temporary and permanent validity where permitted
- Event-specific accreditation
- Venue-specific accreditation
- Multi-venue accreditation
- Temporary event accreditation

- Seasonal accreditation
- Suspension and reactivation
- Permanent revocation
- Automatic expiration
- Expiry monitoring
- Renewal workflows
- Propagation of lifecycle changes to credentials and access rights

Every lifecycle transition shall capture **who performed the action, when it occurred, the reason, previous status and new status.**

Revoked and expired accreditation records shall remain available for authorized reporting and audit purposes.

Matrix Coverage

Board 6 directly addresses:

- 12.1.6 — Expiration Management**
- 12.1.32 — Accreditation Status Workflow**
- 12.1.34 — Accreditation Suspension**
- 12.1.35 — Accreditation Revocation**
- 12.1.36 — Accreditation Expiry Notifications** (*trigger*)
- 12.1.37 — Accreditation Renewal**
- 12.1.38 — Event-Specific Accreditation**
- 12.1.39 — Venue-Specific Accreditation**
- 12.1.40 — Multi-Venue Accreditation**
- 12.1.41 — Temporary Event Accreditation**
- 12.1.42 — Seasonal Accreditation**
- 12.1.45 — Renewal Notifications** (*trigger*)
- 12.1.46 — Expiration Notifications** (*trigger*)

Notification **delivery, templates and communication channels** will be handled centrally in Board 7 rather than duplicated here.

Critical TICVAI behavior

Lifecycle changes should cascade automatically:

Suspension

Accreditation → Suspended → Credential Suspended → Access Disabled

Reactivation

Accreditation → Active → Eligible Credential Reactivated → Access Restored

Expiration

Accreditation → Expired → Credential Expired → Access Terminated

Revocation

Accreditation → Revoked → Credential Revoked → Access Immediately Blocked

This cascade is important to include in the SOW because otherwise development team could implement accreditation status independently from credential/access status, creating a significant security gap.

BOARD 7 — NOTIFICATIONS, COMMUNICATIONS & BULK OPERATIONS

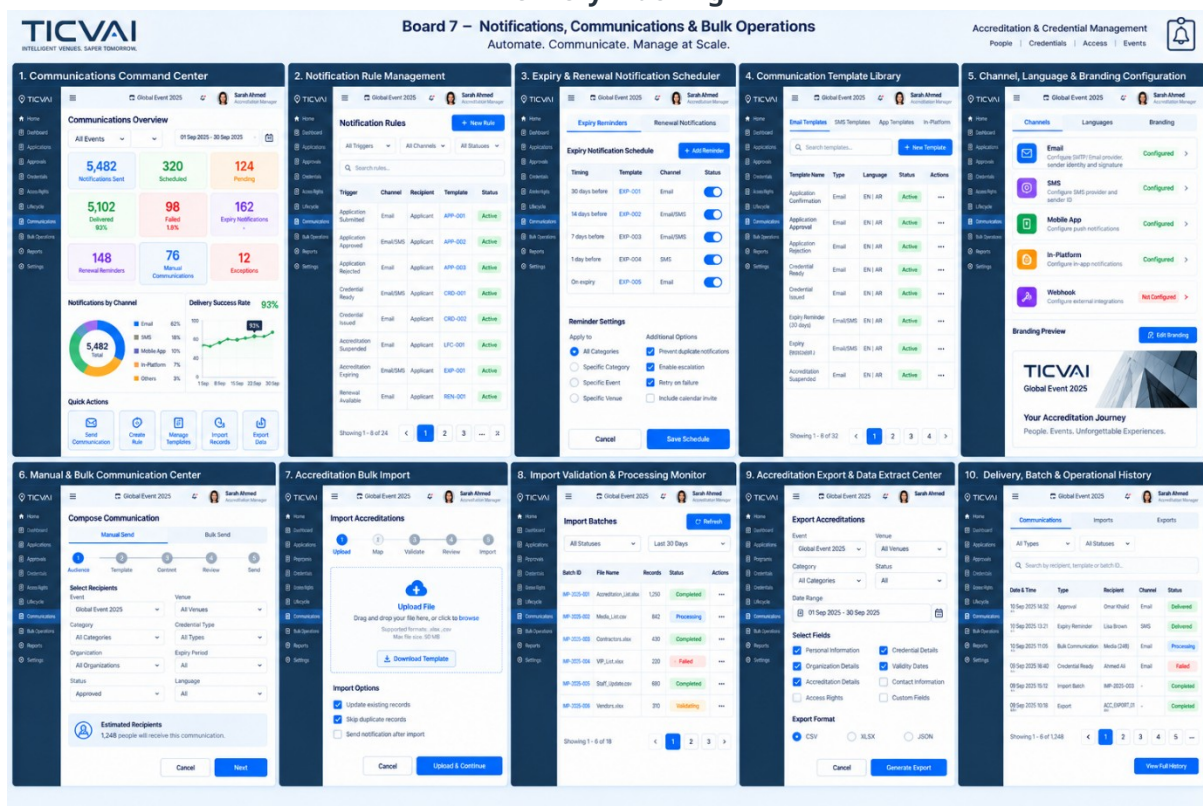
Board Objective

Provide a centralized operational workspace for accreditation communications, notification automation, reusable templates, bulk imports/exports and mass administrative actions.

The solution shall support automated and manual communications throughout the accreditation lifecycle while ensuring that communication rules remain **tenant-, event-, venue-, category- and language-aware**.

The architecture should work as:

Accreditation Event → Trigger → Communication Rule → Template → Channel → Recipient → Delivery Tracking



Screen 1 — Accreditation Communications Command Center

Purpose: Central dashboard for accreditation notifications and operational communications.

Scope of Work

The system shall provide KPI cards including:

- Notifications sent
- Scheduled notifications
- Pending delivery
- Delivered
- Failed

- Approval notifications
- Expiry notifications
- Renewal reminders
- Manual communications
- Communication exceptions

Users shall be able to filter by:

- Tenant
- Event
- Venue
- Accreditation program
- Category
- Notification type
- Channel
- Delivery status
- Date range

Dashboard analytics shall show:

- Delivery success rate
- Notifications by type
- Notifications by channel
- Failure trends
- Upcoming scheduled communications

Quick actions:

Send Communication / Create Rule / Manage Templates / Import Records / Export Data

Key requirements: 12.1.43–12.1.46

Screen 2 — Notification Rule Management

Purpose: Configure when accreditation notifications are automatically triggered.

Scope of Work

Administrators shall be able to create notification rules based on accreditation events.

Supported triggers may include:

- Application submitted
- Additional information required
- Application approved
- Application rejected
- Application resubmitted
- Credential ready

- Credential issued
- Credential activated
- Accreditation suspended
- Accreditation reactivated
- Accreditation revoked
- Accreditation expiring
- Accreditation expired
- Renewal available
- Renewal approved

Each rule shall define:

- Trigger
- Recipient
- Communication channel
- Template
- Timing
- Event/program scope
- Category
- Language
- Active/inactive status

Key requirements: 12.1.43–12.1.46

Screen 3 — Expiry & Renewal Notification Scheduler

Purpose: Configure proactive reminders before accreditation expiry or renewal.

Scope of Work

Administrators shall be able to configure sequences such as:

30 days before expiry → Reminder 1

14 days before expiry → Reminder 2

7 days before expiry → Urgent Reminder

1 day before expiry → Final Reminder

On expiry → Expiration Notice

Renewal schedules may similarly trigger:

- Renewal eligibility notification
- Renewal invitation
- Reminder
- Final reminder
- Renewal confirmation

Administrators shall define:

- Days before/after event
- Recipient
- Template
- Channel
- Escalation rule
- Retry policy

Duplicate notification protection shall prevent unintended repeated messages.

Key requirements: 12.1.36, 12.1.45, 12.1.46

Screen 4 — Communication Template Library

Purpose: Manage reusable accreditation communication templates.

Scope of Work

The system shall provide reusable templates for:

- Application confirmation
- Approval
- Rejection
- Additional information request
- Credential ready
- Credential delivery
- Suspension
- Reactivation
- Revocation
- Expiry reminder
- Expiration
- Renewal invitation
- Renewal approval

Templates shall support configurable variables such as:

- Holder name
- Accreditation ID
- Event
- Venue
- Category
- Validity
- Expiry date
- Credential details
- Collection point

- Renewal link

Templates shall support versioning and active/inactive states.

Key requirements: 12.1.43–12.1.46, 12.1.56

Screen 5 — Channel, Language & Branding Configuration

Purpose: Control how communications are delivered and branded.

Scope of Work

The system shall support configurable channels such as:

- Email
- SMS
- Mobile/app notification
- In-platform notification
- External messaging provider where integrated

Administrators shall configure:

- Sender identity
- Reply information
- Default language
- Alternative languages
- Tenant logo
- Event branding
- Venue branding
- Header/footer
- Contact information

Templates may inherit branding from the tenant and override it for individual programs/events.

The system shall support multilingual template variants where configured.

Key requirement: 12.1.57

Screen 6 — Manual & Bulk Communication Center

Purpose: Allow authorized operators to communicate with selected accreditation populations.

Scope of Work

Users shall be able to select recipients by:

- Accreditation category
- Organization
- Event
- Venue
- Program

- Status
- Expiry period
- Credential type

Examples:

Send collection instructions to all approved media.

Notify all contractors that accreditation collection closes at 18:00.

Before sending, the system shall display:

- Recipient count
- Channel
- Template
- Language
- Preview
- Estimated delivery volume

Actions:

Save Draft / Preview / Schedule / Send Now

Bulk communications shall be fully auditable.

Screen 7 — Accreditation Bulk Import

Purpose: Import accreditation records at scale.

Scope of Work

The system shall support bulk import using approved structured file formats.

Administrators shall be able to:

- Download import template
- Upload file
- Map columns
- Validate records
- Detect duplicates
- Preview changes
- Resolve errors
- Submit import

Supported import data may include:

- Holder information
- Organization
- Accreditation category
- Event
- Venue

- Validity
- Credential information where permitted

The system shall provide:

Valid / Warning / Error / Duplicate

status for each imported row.

No records shall be committed until validation is complete.

Key requirement: 12.1.54

Screen 8 — Import Validation & Processing Monitor

Purpose: Govern and monitor bulk accreditation imports.

Scope of Work

The screen shall display:

- Import batch ID
- File name
- Uploaded by
- Upload date/time
- Total records
- Valid records
- Warning records
- Failed records
- Duplicate records
- Processing status

Users shall be able to inspect row-level errors and download an error report.

Processing statuses shall include:

Uploaded → Validating → Ready → Processing → Completed / Partially Completed / Failed

Every imported record shall maintain the originating batch reference for auditability.

Screen 9 — Accreditation Export & Data Extract Center

Purpose: Export authorized accreditation data for operational or reporting purposes.

Scope of Work

Authorized users shall be able to configure exports based on:

- Program
- Event
- Venue
- Category
- Organization

- Status
- Validity
- Credential type
- Date range

Users shall select permitted data fields.

Export formats may include:

- CSV
- XLSX
- Other configured machine-readable formats

Sensitive fields shall be controlled through RBAC.

The system shall capture:

- Requested by
- Export date/time
- Filters
- Fields exported
- Number of records
- Export status

Large exports may be processed asynchronously and made available securely to the requesting user.

Key requirement: 12.1.55

Screen 10 — Delivery, Batch & Operational History

Purpose: Provide a consolidated history of notifications, communications, imports and exports.

Scope of Work

The screen shall maintain an operational history covering:

Communications

- Recipient
- Notification type
- Template
- Channel
- Sent time
- Delivery status
- Failure reason
- Retry history

Imports

- Batch
- User

- File
- Records processed
- Success/failure

Exports

- Export request
- User
- Scope
- Record count
- Completion status

Users shall be able to search and filter historical operations.

Authorized administrators shall be able to retry failed communications without recreating the original transaction.

Board 7 Acceptance Criteria

Board 7 shall be considered complete when TICVAI can centrally manage accreditation communications and high-volume administrative operations without requiring manual communication outside the platform.

The solution must support:

- Event-driven accreditation notifications
- Approval notifications
- Expiry notifications
- Renewal notifications
- Rejection and lifecycle communications
- Configurable communication templates
- Multilingual templates
- Tenant/event branding
- Multiple communication channels
- Manual communications
- Bulk communications
- Accreditation bulk import
- Import validation
- Duplicate detection during import
- Accreditation export
- Communication delivery tracking
- Batch-operation history

All communication and bulk-operation activities shall be subject to **RBAC, tenant segregation and audit logging**.

Matrix Coverage

Board 7 directly addresses:

12.1.36 — Accreditation Expiry Notifications

12.1.43 — Accreditation Notifications

12.1.44 — Approval Notifications

12.1.45 — Renewal Notifications

12.1.46 — Expiration Notifications

12.1.54 — Accreditation Import

12.1.55 — Accreditation Export

12.1.56 — Accreditation Templates

12.1.57 — Accreditation Branding

The final **Board 8** should then consolidate the remaining enterprise capabilities: **Accreditation Dashboard & Reporting, Utilization Analytics, Access Reporting, Trend Analysis, API, Webhooks, Access Audit Trail and Immutable Audit Logs**. That will close the entire 12.1.1–12.1.58 matrix without leaving any significant requirement uncovered.

BOARD 8 — ACCREDITATION ANALYTICS, REPORTING, INTEGRATION & AUDIT

Board Objective

Provide the enterprise oversight layer for Accreditation & Credential Management, enabling management teams to monitor accreditation performance, analyze utilization and access activity, investigate historical actions, and integrate accreditation services with external platforms through secure APIs and webhooks.

This board shall consolidate:

Dashboard → Reporting → Utilization → Access Activity → Trends → Audit → API → Webhooks → Integration Monitoring



Screen 1 — Accreditation Executive Dashboard

Purpose: Provide management with a real-time overview of the complete accreditation operation.

Scope of Work

The system shall provide configurable KPI cards including:

- Total accreditation holders
- Active accreditations
- Pending applications
- Approved
- Rejected
- Suspended

- Revoked
- Expired
- Expiring soon
- Credentials issued
- Active credentials
- Accreditation utilization rate

Dashboard filters shall include:

- Tenant
- Event
- Venue
- Accreditation program
- Category
- Organization
- Date range

Visualizations should include:

- Accreditation by status
- Accreditation by category
- Accreditation by venue
- Accreditation by organization
- Application-to-approval funnel
- Credential issuance status

Users shall be able to drill from dashboard KPIs into underlying records.

Key requirements: 12.1.7, 12.1.47

Screen 2 — Accreditation Status & Portfolio Reporting

Purpose: Report on active, expired, suspended and revoked accreditation records.

Scope of Work

The system shall provide configurable accreditation portfolio reports.

Users shall be able to analyze:

- Active accreditations
- Expired accreditations
- Revoked accreditations
- Suspended accreditations
- Renewal pending
- Temporary accreditation
- Seasonal accreditation

- Event-specific accreditation
- Venue-specific accreditation
- Multi-venue accreditation

Reports shall support grouping by:

- Category
- Event
- Venue
- Organization
- Program
- Accreditation type

Users shall be able to drill into individual accreditation records.

Key requirement: 12.1.7

Screen 3 — Accreditation Utilization Analytics

Purpose: Determine whether issued accreditations are actually being used.

Scope of Work

The system shall analyze credential/access activity against issued accreditation records.

Metrics shall include:

- Accreditations issued
- Accreditations activated
- Accreditations used
- Never-used accreditations
- First-use date/time
- Last-use date/time
- Number of access transactions
- Utilization percentage
- Utilization by category
- Utilization by venue
- Utilization by event

Example:

1,250 Media Accreditations Issued

1,086 Activated

972 Used

278 Never Used

77.8% Utilization

This information shall use access-event data from the TICVAI Access Control layer rather than duplicating validation logic.

Key requirement: 12.1.48

Screen 4 — Accreditation Access Activity Reporting

Purpose: Provide detailed reporting of accreditation access activity.

Scope of Work

The system shall display access transactions including:

- Holder
- Accreditation ID
- Credential ID
- Category
- Event
- Venue
- Zone/area
- Access point
- Date/time
- Access result
- Reason/result code

Users shall be able to filter:

Granted / Denied / Exception / Override

The report shall support investigation of:

- Unusual access
- Repeated denied attempts
- Restricted-area attempts
- Credential misuse
- Access outside schedule
- Suspended credential attempts

Where available, records shall link back to the original Access Control transaction.

Key requirements: 12.1.31, 12.1.49

Screen 5 — Accreditation Trend & Comparative Analysis

Purpose: Analyze accreditation behavior over time.

Scope of Work

The system shall provide trend reporting for:

- Application volume
- Approval rate
- Rejection rate
- Average approval time

- Credential issuance
- Accreditation utilization
- Suspension rate
- Revocation rate
- Renewal rate
- Expiration volume
- Access activity

Users shall compare:

- Event vs event
- Venue vs venue
- Category vs category
- Period vs period
- Organization vs organization

Trend charts shall support daily, weekly, monthly, seasonal and event-level analysis.

Key requirement: 12.1.50

Screen 6 — Accreditation Audit Reporting

Purpose: Provide compliance and management reporting across accreditation actions.

Scope of Work

Authorized users shall be able to report on:

- Application creation
- Profile modifications
- Identity verification
- Document verification
- Approval/rejection
- Credential issuance
- Badge reprinting
- Access assignment
- Temporary access grants
- Suspension
- Reactivation
- Revocation
- Renewal
- Imports
- Exports
- Administrative configuration changes

Audit reports shall support:

- User
- Action
- Date/time
- Entity
- Event
- Previous value
- New value
- Source
- Reason

Key requirement: 12.1.51

Screen 7 — Immutable Accreditation Audit Log

Purpose: Maintain the authoritative security record of all accreditation actions.

Scope of Work

The system shall maintain immutable audit records for security-sensitive accreditation transactions.

Audit records shall capture:

- Audit ID
- Timestamp
- User/service identity
- Tenant
- Action
- Entity type
- Entity ID
- Previous value
- New value
- Source application/API
- Correlation/reference ID
- Result
- Reason where applicable

Standard users shall **not be able to edit or delete audit records**.

Retention shall follow configured tenant/compliance policy.

Authorized users shall be able to search and export permitted audit records.

Key requirement: 12.1.58

Screen 8 — Accreditation API Management

Purpose: Expose accreditation functionality securely to approved external systems.

Scope of Work

The system shall expose governed APIs for functions such as:

- Create accreditation application
- Retrieve application
- Retrieve holder
- Update permitted profile information
- Retrieve accreditation status
- Retrieve credential status
- Retrieve access assignment
- Create/import accreditation records
- Suspend accreditation
- Revoke accreditation
- Retrieve reports where authorized

API management shall support:

- Authentication
- Authorization/scopes
- Tenant isolation
- Rate limits
- API version
- Request/response logging
- Error handling

API access shall comply with TICVAI's central integration architecture.

Key requirement: 12.1.52

Screen 9 — Accreditation Webhook Management

Purpose: Notify external systems when accreditation events occur.

Scope of Work

Administrators shall be able to configure webhook subscriptions for events such as:

- Application submitted
- Application approved
- Application rejected
- Accreditation activated
- Credential issued
- Credential activated

- Accreditation suspended
- Accreditation revoked
- Accreditation expired
- Accreditation renewed
- Access profile changed

Webhook configuration shall include:

- Endpoint
- Event subscriptions
- Authentication/security
- Secret/signature
- Retry policy
- Active/inactive
- Delivery history

Failed deliveries shall support controlled retries.

Key requirement: 12.1.53

Screen 10 — Integration & Data Exchange Monitor

Purpose: Monitor accreditation APIs, webhooks and external-system synchronization.

Scope of Work

The screen shall provide operational visibility into:

- API requests
- Successful transactions
- Failed API transactions
- Webhook deliveries
- Failed webhooks
- Retry queue
- Access Control synchronization
- Credential synchronization
- Import/export jobs
- External identity-verification integrations

Administrators shall be able to inspect:

- Transaction ID
- Integration
- Operation
- Date/time
- Status

- Response/result
- Retry count
- Error details

Quick actions:

Retry / View Payload Metadata / View Error / Open Audit Record

Sensitive payload information shall be masked according to security policy.

Board 8 Acceptance Criteria

Board 8 shall be considered complete when TICVAI provides management, operations, compliance teams and authorized external systems with a comprehensive view of accreditation activity.

The solution shall provide:

- Accreditation executive dashboard
- Active/expired/revoked reporting
- Utilization analytics
- Access-activity reporting
- Trend analysis
- Audit reporting
- Immutable audit logs
- Accreditation APIs
- Accreditation webhooks
- API/webhook operational monitoring
- Data export capability
- Integration with Access Control reporting data

Reporting shall respect **tenant segregation, venue/event permissions and RBAC**.

Audit records shall remain immutable to normal application users.

Matrix Coverage

Board 8 directly addresses:

12.1.7 — Accreditation Reporting

12.1.31 — Access Audit Trail

12.1.47 — Accreditation Dashboard

12.1.48 — Accreditation Utilization Reporting

12.1.49 — Accreditation Access Reporting

12.1.50 — Accreditation Trend Analysis

12.1.51 — Accreditation Audit Reporting

12.1.52 — Accreditation API

12.1.53 — Accreditation Webhooks

12.1.58 — Accreditation Audit Logs

It also consolidates reporting information generated by Boards 1–7.

Final Accreditation Architecture

With Board 8, the complete module becomes:

Board	Capability
1	Command Center, Programs & Registration
2	Holder Profile, Documents & Identity
3	Approval Workflow & Governance
4	Credential Issuance, Media & Badges
5	Access Rights, Zones & Schedules
6	Lifecycle, Validity, Expiry & Renewal
7	Notifications, Communications & Bulk Operations
8	Analytics, Reporting, API, Webhooks & Audit

My recommendation is to stop at Board 8. We now have a logical end-to-end accreditation architecture, and adding Board 9 would mostly start duplicating reporting, Access Control, integration, or platform-level administration already covered elsewhere in TICVAI.